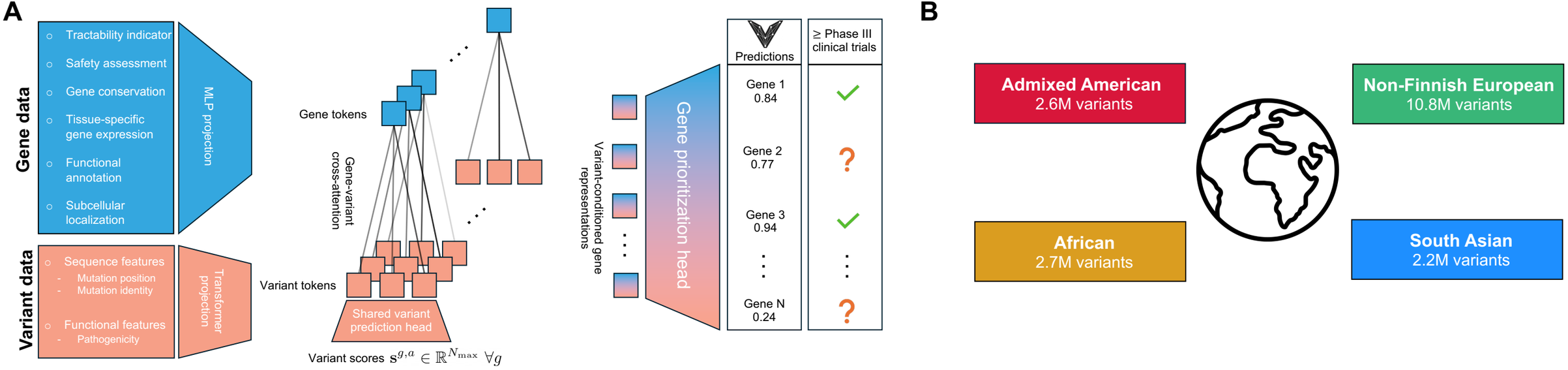
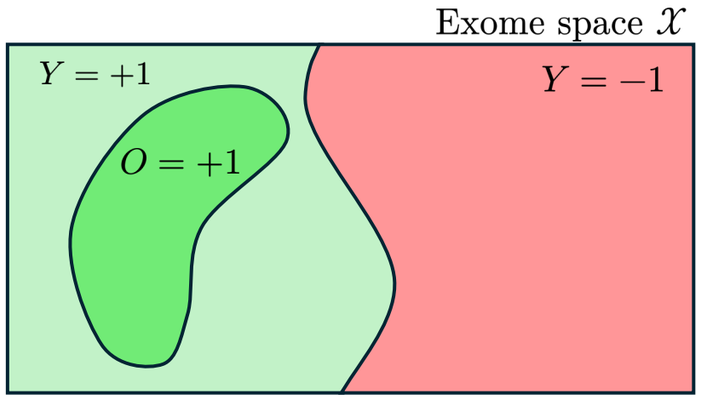
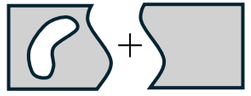


C





$= \text{Unlabelled targets } \mathcal{U}$

$$\mathcal{L}_{\text{gene}} = \underbrace{-\pi \mathbb{E}_p^{\text{obs}}[\log S_\theta(\mathbf{X})]}_{\text{Observed positives } (O=+1)} + \underbrace{\max\left(0, \underbrace{\pi \mathbb{E}_p^{\text{obs}}[\log(1 - S_\theta(\mathbf{X}))]}_{\text{Observed positives } (O=+1)} - \underbrace{\mathbb{E}_u[\log(1 - S_\theta(\mathbf{X}))]}_{\text{Unlabelled pool } (\mathcal{U})}\right)}_{\text{Non-negative risk estimator (negative class)}}$$

$$\mathcal{L}_{\text{variant}} = -\frac{1}{|\mathcal{M}|} \sum_{(g,i)\in\mathcal{M}} [y_i^g \log \sigma(s_i^{g,a}) + (1 - y_i^g) \log(1 - \sigma(s_i^{g,a}))]$$

$$\boxed{\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{gene}} + \lambda_{\text{v}} \mathcal{L}_{\text{variant}}}$$